

Mission Advantage Anywhere

Contingency communications for the first 72 hours.

Product

Mission Advantage Anywhere

Builder

Merge Combinator, LLC

CAGE

11TF7

Status

Fielding, unclassified baseline

Node classes

Mobile · Flyaway · Persist

Issued

August 2026

Mission Advantage Anywhere is a family of carry-on kits that keeps a dispersed team connected and reporting when the network they normally use is gone. It moves traffic over whatever path is available, whether mesh, satellite or cellular, and puts the team's position, status and observations onto the command and control picture the unit already watches.

01 Product overview

Mission Advantage Anywhere is a deployable communications and reporting kit for continuity of operations while the mission and the network adapt. Teams disperse to austere and contingency locations, relocate a command post, or operate outside the base network boundary, and for the first 24 to 72 hours they are limited to what one person can carry. The kit is what that person carries.

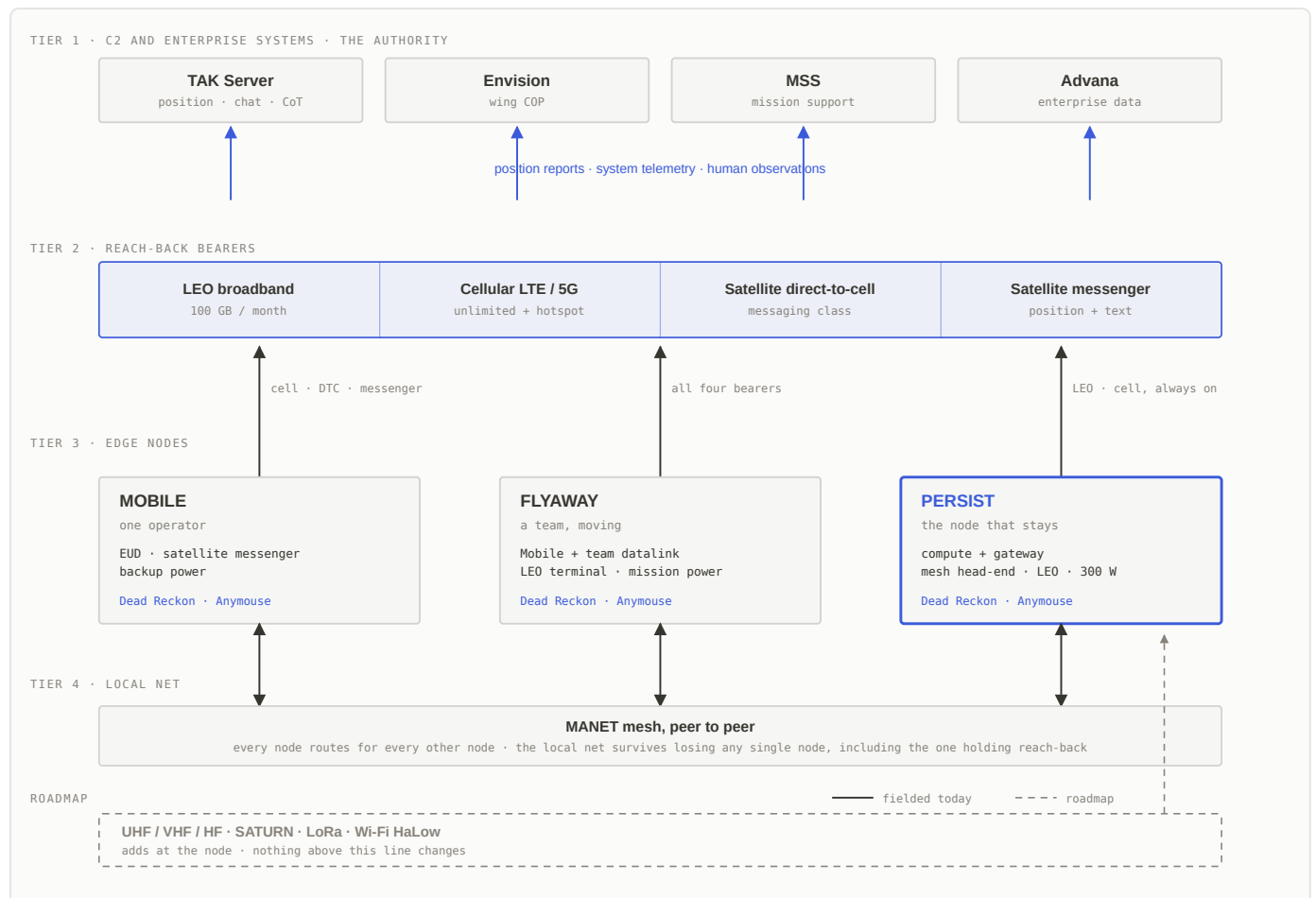
It is delivered in three node classes that share one software layer and one data model. A Mobile node puts a single operator on the picture. A Flyaway node moves a team with a local datalink and satellite reach-back. A Persist node stays and becomes the head-end for the local net. Any combination of them forms a working network, and every node routes for every other node, so the net survives losing any one of them, including the one holding reach-back.

The kit is additive by design. Existing systems remain the authority. The kit carries traffic to them and reports the mission around itself while it does.

02 Key features

- **Three node classes, one software layer.** Mobile, Flyaway and Persist scale from one operator to a standing site without changing how data reaches the C2 picture.
- **PACE is part of the design.** Primary, alternate, contingency and emergency paths are set before the mission. The kit reports which path is carrying traffic and when that changed.
- **Peer-to-peer local net.** Every node routes for every other node. No hub to lose.
- **It reports itself.** System and mission telemetry through Dead Reckon; human observation through Anymouse. The kit is instrumented, so exercises produce data instead of anecdote.
- **Rides the systems already in place.** Position and traffic land in TAK; telemetry and reporting feed Envision, MSS and Advana.
- **Commercial and replaceable.** Components are commercial off the shelf and individually substitutable. No single vendor is load-bearing.
- **Carried, not operated, in flight.** Devices stay powered off in the cockpit and all use is off-aircraft. No cockpit-use approval is required and nothing in the design depends on one.
- **An accreditation path, stated honestly.** Components have IL4 precedent and the managed service supports CUI today. The specific combination is new and carries its own review.

03 System view · SV-2



SV-2 · Systems resource flow. Modularity is vertical and connectivity is horizontal. Any node class reaches any bearer it carries, and any bearer reaches the same four enterprise systems, so adding a node or a waveform changes one tier without touching the others. The dashed line is the claim worth testing: roadmap bearers integrate at the node, and nothing above the node has to change to accept them.

04 Node classes



Mobile

Pelican 1120 charge case. iPhone 17 Pro end user device, Garmin inReach Mini 3 Plus satellite messenger, backup power and cabling. Charges closed; issued ready to use.



Flyaway

Single-person carry. The Mobile case plus a Starlink Mini LEO terminal, power adapter and team datalink, packed to move.

Photography shows indicative configurations. Contents vary by mission and by the options selected at order. Persist is configured to order and is not pictured.

Node	What it is	Carries	Bearers	ROM
Mobile	One operator on the picture. Provisioned and ready to issue.	Pelican 1120 charge case, iPhone 17 Pro, Garmin inReach Mini 3 Plus, backup power, cabling	Cellular, direct-to-cell, satellite messenger	\$2,500
Flyaway	A team or vehicle, moving. Single-person carry.	Mobile node plus Starlink Mini LEO terminal, power adapter, team datalink, tactical pack	All four, plus local mesh	\$6,000
Persist	The node that stays. Command communications centre in a single-person carry-on. Configured to order.	Compute and gateway, encrypted mesh head-end for the local dark net, LEO reach-back, high-capacity power	All bearers, head-end for the local net	\$30,000

05 General specifications

Kit figures are the sum of the components in the baseline configuration. Component figures are manufacturer published.

Specification	Mobile	Flyaway	Persist
Form factor	Pelican 1120 Protector Case	Mobile case and LEO terminal in a tactical pack, single-person carry	Rugged carry-on case
Dimensions	21.4 × 17.2 × 9.8 cm exterior 8.41 × 6.76 × 3.87 in	Driven by the 29.9 × 25.9 × 3.9 cm LEO panel and the Mobile case. Fits a 25 to 35 L pack	56 × 36 × 23 cm 22 × 14 × 9 in, carry-on
Weight	0.98 kg (2.2 lb) case, device, messenger	2.1 kg (4.6 lb) excluding pack Mobile plus LEO terminal	11 to 25 kg (25 to 55 lb) by configuration
Power	Internal battery and power bank, USB-C. Charges with the case closed	Mobile power plus 25 to 40 W average for the LEO terminal, 12 to 48 V input	300 W bus. 12 to 36 VDC, 90 to 264 VAC, vehicle and AC input
Endurance	Messenger runs multi-day at standard reporting intervals on 1800 mAh. The case charges both devices closed	A 99 Wh battery supports roughly 2.5 to 4 hours of terminal-on time. Sized per mission	Accepts BB-2590 and BB-2557 batteries. Endurance scales with battery count
Operating temp	Messenger -20 °C to 60 °C. End user device 0 °C to 35 °C per manufacturer	LEO terminal -30 °C to 50 °C. Mobile limits apply to the devices inside	-20 °C to 50 °C
Ingress rating	Case IP67. Messenger IP67	LEO terminal IP67 with power cable installed. Mobile case IP67	Watertight when closed, IP67 sealed case
Aircraft carriage	Carried, not operated. Devices powered off in flight; all use off-aircraft. F-22 stowage envelope is a dimensional and mass question only. No EMI or airworthiness qualification is in scope.		
Set-up time	Measured from case open to first report on the C2 picture, and published from the first rehearsal. This is one of the two numbers the programme reports.		

Components and mass

Published figures for the commercial items in the baseline configuration. Each is individually substitutable.

Component	Role	Dimensions	Mass	Rating
Pelican 1120	Charge case, Mobile node	21.4 × 17.2 × 9.8 cm	0.65 kg	IP67, crushproof, watertight
iPhone 17 Pro	End user device	15.0 × 7.19 × 0.88 cm	0.21 kg	IP68, 0 °C to 35 °C operating
Garmin inReach Mini 3 Plus	Satellite messenger	10.0 × 5.2 × 2.6 cm	0.12 kg	IP67, -20 °C to 60 °C, 1800 mAh
Starlink Mini	LEO broadband terminal	29.9 × 25.9 × 3.9 cm	1.10 kg	IP67, -30 °C to 50 °C, 25 to 40 W

Power bank, adapter, cabling and pack are selected per mission and are not included in the kit mass figures above.

06 Applications

Command post continuity of operations

The kit gives a relocating command post a working net and a reporting path before permanent communications exist.

Use case: A wing command post displaces to an alternate site. A Persist node establishes the local net and reach-back within the first hours; Mobile nodes keep dispersed staff on the picture while the primary path is rebuilt.

Dispersed beddown, first 72 hours

Aircrew and maintainers arrive at a contingency location with what one person can carry and still need to report.

Use case: A team beds down at an austere airfield outside the base network boundary. Flyaway nodes carry team traffic locally and push status and position to the C2 picture until the site is connected properly.

Flight line coordination and notification

Where coordination between stakeholders runs on personal cell phones and legacy voice, the kit closes the notification gap.

Use case: An event on the flight line requires an immediate coordinated response across organisations that do not share a net. The kit puts the same picture in front of all of them instead of relying on a chain of phone calls.

Personnel and asset accountability

Position and status reporting from people and equipment that would otherwise be tracked by spreadsheet and phone call.

Use case: During a dispersal or a domestic response, tracking of aircraft, equipment and personnel is currently manual and degrades under stress. Nodes report continuously into the systems the command post already uses.

Exercise instrumentation

The kit records what happened, so a rehearsal produces measured data rather than recollection.

Use case: During a major exercise, Dead Reckon records path state and system events while Anymouse captures what operators observed. The after-action product is a timeline, not a memory.

Defence support to civil authorities

A commercial, unclassified kit that works alongside civil agencies without requiring them to join a military network.

Use case: During a wildfire or storm response, guard teams operating with state and county partners share position and status on an unclassified picture that all participants can see.

07 Data and reporting

The kit carries traffic, and while it does, it reports the mission around itself. Two software components ride on every node class.

SOFTWARE COMPONENT

Dead Reckon

System and mission telemetry into the C2 picture. Records path state, node health and system events, and fills the integration gaps where existing tablet plug-ins do not yet reach.

SOFTWARE COMPONENT

Anymouse

Human context. Named after the anonymous aviation safety report box and built for the same reason: to capture the observation that would otherwise be lost.

Where the data lands

System	What reaches it
TAK	Team traffic, position reports, chat
Envision	Wing common operating picture, status and field reporting
MSS	Mission support reporting
Advana	Enterprise data and analysis

08 Integration list

Current and planned component integrations. Components are commercial off the shelf and individually substitutable; no single supplier is load-bearing.

COMPUTE AND END USER DEVICE

- Provisioned mobile end user device
- Tablet end user device, optional
- Persist compute and gateway module

DATALINK AND RADIOS

- Encrypted MANET mesh head-end **NOW**
- Baseline team datalink **NOW**
- LoRa, Wi-Fi HaLow **NEXT**
- UHF / VHF / HF, SATURN **NEXT**

SATELLITE AND CELLULAR

- LEO broadband terminal **NOW**
- Commercial cellular, rotating SIM identifiers **NOW**
- Satellite direct-to-cell messaging **NOW**
- Satellite messenger, position and text **NOW**

SOFTWARE

- Dead Reckon telemetry and integration
- Anymouse observation reporting
- TAK client and server integration

POWER

- Integrated charge case, backup power
- Common-battery mission power module
- Vehicle and AC input on Persist

POSITION, NAVIGATION AND TIMING

- Alternate PNT node **NEXT**
- Low probability of intercept paging **NEXT**
- Asset tracking tags **NEXT**

09 Security and accreditation posture

Stated plainly, because the answer is not "already certified" and a partner deserves to know that before an exercise rather than during one.

- **Components.** The kit adapts established IL4 components. Several are under active review by the USAF operational test community toward IL-6, and the managed service supports CUI today.
- **The combination is new.** This specific hardware and software combination, in this use case and configuration, has not been reviewed or approved before. No authorisation is inherited and none is claimed.
- **Device management.** Devices enrol under an existing electronic flight bag mobile device management programme rather than a new one.
- **Spectrum.** Frequency authorisation for the datalink is an open item and a named engineering task, not an assumption.
- **Classified operation.** The baseline is unclassified. SECRET operation is scoped and priced separately after discovery. Start unclassified, arrive classified.

10 Value proposition

Why Mission Advantage Anywhere. Communications planning fails in a predictable place: the stretch between arriving somewhere and being connected there. Dispersal, mission command under degraded conditions and distributed control all assume that stretch has already been covered. Covering it in a form one person can carry onto an aircraft is the job this kit does.

It bridges the gap

Continuity of operations while the mission and the network adapt, then a clean handover to whatever the site is eventually wired with.

It is additive

No rip and replace, no new command and control authority. Existing systems stay the authority and the kit feeds them.

It produces evidence

Two measured numbers: time to restore the net when the primary path drops, and time from an edge observation to its appearance on the C2 picture.

It starts small

A single node is a real capability and a real test. Scaling from there is a quantity decision.

11 Target customers

PRIMARY

- Air National Guard wings and wing command posts
- Air defence and homeland defence units
- Units operating under agile combat employment concepts

SECONDARY

- Joint partners requiring cross-service reporting at the edge
- State emergency management and domestic response
- Integrators and primes needing an edge reporting layer under an existing programme

12 Talking points

Short pitch. Mission Advantage Anywhere is a carry-on communications kit for continuity of operations while the mission and the network adapt. It gives a dispersed team a local net and a reporting path in minutes, over whatever bearer is available, and lands their position and observations on the C2 picture the unit already watches. Existing systems stay the authority, and the kit feeds them.

1. **The gap is real and it is measurable.** The first 24 to 72 hours of a beddown, and the coordination delay at home station, are both instrumented problems, not opinions.
2. **A detection that reaches nobody is not a detection.** Sensing is the easiest thing to buy and the least useful thing to buy first.
3. **Additive by design.** The kit claims no authority and replaces no system, which makes it approvable in months rather than years.
4. **Start with one node.** The entry configuration is a real capability at a price that fits inside a routine purchase.
5. **Instrumented by default.** Every exercise produces data. That data is what supports a documented requirement later.
6. **Commercial and replaceable.** Components can be substituted without redesigning the system, so obsolescence is a purchase, not a programme.
7. **An honest accreditation story.** We say what is approved, what is not, and what the path is. That is the part most vendors leave out.

One-liners

The network shows up in a backpack.

It carries the traffic, and it reports the mission around it.

Existing systems stay the authority. We feed them.

A detection that reaches nobody is not a detection.

Use it when the beddown is in twelve hours and the network is in thirty days.

It is how the edge reaches the C2 system you already have.

Every node routes for every other node. There is no hub to lose.

13 What it is not

- Not a command and control system, and not a new C2 authority.
- Not a counter-UAS system. It does not detect and it does not defeat.
- Not a replacement for tactical radios or for a fielded programme of record.
- Not accredited for classified operation today.
- Not certified for cockpit use. It is carried powered off and used off-aircraft.
- Not a promise of automatic failover. Path state is reported; automated failover is an engineering objective and will be described by measured behaviour.